

Scalable Visual Foundation Models for Remote Sensing, Multimodal Agricultural AI, and Calibrated Spatial Reasoning

Postdoctoral Research Plan and Statement of Research Interests

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1. Research Vision: Foundation Models for African Remote Sensing & Agriculture

Sustainable agriculture and food security across the African continent face severe operational challenges exacerbated by climate volatility, soil degradation, and emerging crop pathologies. Satellite constellations (e.g., Sentinel-2, Landsat) and low-altitude drone visual feeds provide unprecedented spatial and multi-spectral observations. However, transforming these multi-sensor feeds into actionable agronomic insights is critically bottlenecked by the **supervision gap**: standard supervised deep learning models fail to generalize across diverse African agro-ecological zones and require massive, manually annotated pixel-level masks that do not exist for indigenous farming systems.

Brute-force end-to-end retraining risks catastrophic forgetting and severe distribution collapse. The frontier of robust visual intelligence lies in **inference-time foundation-model composition** ($\Delta\theta = 0$), **multi-view geometric consensus**, and **physically grounded multimodal reasoning**. My doctoral research has developed theoretical and practical frameworks for zero-shot spatial segmentation, distribution-preserving semantic retention, and self-supervised reinforcement learning process rewards. At the **College of Computing (COLCOM)** at **Mohammed VI Polytechnic University (UM6P)**, I plan to apply these foundations to establish continental-scale, calibrated AI systems for remote sensing, precision crop monitoring, and automated agricultural knowledge discovery, powered by the **TOUBKAL** supercomputing infrastructure.

2. Methodological Foundations from Doctoral Research

My proposed program builds upon three rigorous methodological pillars established during my PhD:

- **Training-Free Open-Vocabulary Spatial Perception (ACCV 2026, WACV 2027)**: Reconciled 2D promptable segmenters (SAM 2.1) with dense vision-language embeddings using projective camera geometry (K, T), ray casting, and spatial multi-view consensus. Without updating underlying model parameters ($\Delta\theta = 0$), this framework achieved state-of-the-art harmonic mean accuracy across massive spatial benchmarks ($\sim 50M$ points, 312 scenes).
- **Preventing Premature Semantic Collapse (ICLR 2027 Submission)**: Identified and formalized how greedy argmax decisions in foundation model composition permanently discard rare candidate semantics. Proved that retaining compact top- k distributions enables downstream geometric consensus to recover subtle novel categories (+6.40 mIoU boost), crucial for detecting rare crop diseases or micro-parcels.
- **Self-Supervised Multimodal Reasoning via Process Rewards (SACAIR 2026 Submission)**: Formulated physical cross-view agreement across independent observations as an intrinsic, label-free process reward for Vision-Language Models (**Qwen2.5-VL-7B**). Implemented online Group Relative Policy Optimization (**GRPO/VERL**) on **vLLM**, eliminating area-bias specification gaming ($\rho > 0.72 \rightarrow 0.334$) and spatial hallucinations without human reward labels.
- **Reliability Calibration and Uncertainty Quantification (Target: IEEE TPAMI)**: Developed **RCASA**, proving class-wise bounded updates and probability simplex preservation by uniting Platt scaling with Bernoulli uncertainty. This guarantees safe, calibrated open-set detection under severe out-of-distribution shifts.

3. Proposed Postdoctoral Research Program at UM6P (COLCOM)

Leveraging UM6P's high-performance computing resources, I propose a three-thrust research program tailored to COLCOM's strategic focus on agriculture, remote sensing, and multimodal AI:

Thrust 1: Open-Vocabulary Multi-Sensor Perception for Satellite and Drone Monitoring

Agricultural remote sensing requires fusing heterogeneous visual modalities—coarse-resolution multi-spectral satellite imagery with high-resolution RGB/multispectral drone surveys. I will develop:

- **Cross-Sensor Spatial Lifting:** Formulating projective coordinate mappings and terrain elevation models to lift multi-temporal 2D drone and satellite views into unified spatial feature representations without ground-truth terrain annotations.
- **Zero-Shot Crop Canopy & Parcel Delineation:** Adapting promptable foundation models (SAM 2.1) with open-vocabulary text embeddings (CLIP/OpenCLIP) to segment agricultural field boundaries, crop types, and canopy coverage under zero-shot and few-shot conditions.
- **Early Stress & Pathology Anomaly Detection:** Deploying calibration bounds (*RCASA*) to identify subtle spectral anomalies (water stress, nutrient deficiency, pest infestation) with rigorous false-positive control.

Thrust 2: Physically Grounded Multimodal Agricultural Reasoning Agents

A key goal of COLCOM is the development of LLM and VLM systems for agricultural knowledge and decision support. Today's commercial VLMs frequently hallucinate agronomic diagnoses when prompted with field imagery. I will address this by:

- **Multi-Angle Process Rewards for Agronomic Grounding:** Extending my SACAIR 2026 GRPO reinforcement learning pipeline to agricultural reasoning: rewarding the VLM when its textual deductions (e.g., leaf lesion diagnosis, crop stage estimation) remain consistent across multi-view drone captures and multi-temporal satellite passes.
- **Open-Source Agricultural Knowledge Distillation:** Structuring verified multimodal reasoning traces into open-source African agricultural knowledge graphs, connecting visual observations with local soil, weather, and agronomic guidelines.

Thrust 3: High-Throughput Scaling on TOUBKAL & Open-Source African Benchmarks

Conducting continental-scale agricultural analysis requires industrial-grade computing pipelines:

- **Distributed TOUBKAL Pipeline:** Scaling foundation-model inference, ray casting, and online RL rollouts across the multi-node GPU architecture of the **African Supercomputing Center (ASCC)**. Utilizing float16/bfloat16 activation caching and chunked processing to evaluate multi-terabyte satellite datasets efficiently.
- **Open-Source African Remote Sensing Benchmarks:** Curating and publicly releasing standardized, multi-modal African agricultural benchmarks (drone imagery, satellite time-series, and expert agronomic annotations) to establish COLCOM as the premier global benchmark hub for low-supervision agricultural AI.

4. Expected Scientific Impact, Mentorship & Pan-African Leadership

- **Scientific Dissemination:** Targeting flagship international venues: **CVPR, ICCV, ECCV, ICLR, NeurIPS** in computer vision and machine learning, and **IEEE TPAMI, IEEE TGRS** for comprehensive journal monographs.
- **Student Mentorship:** Actively co-supervising PhD and master's students at COLCOM, fostering rigorous mathematical grounding, reproducible systems engineering, and effective paper writing.
- **Collaborative Proposals:** Partnering with COLCOM faculty on international grant applications (e.g., Horizon Europe, CGIAR, and African Union AI initiatives) to establish sustained funding pipelines.
- **Pan-African Leadership:** As a researcher transitioning from Ghana's premier institution to Morocco's flagship technological university, I will champion South-South research exchanges, bridging West and North African computing talents.